

INDUSTRIAL AUTOMATION TRAINING ON PLC, SCADA & VFD






Course Overview:

This training is tailor made to equip operations personnel with essential knowledge and skills to understand the basics of industrial automation in PLC, SCADA and VFD systems.

Target group:

This course is suitable for all those who actively or passively interact with industrial control systems in their day to day work.

Learning outcomes:



Upon completion of this module, participants will be able to:

- Understand PLC components and inputs/outputs wiring.
- Understand sensors and switches configuration in connection to PLCs.
- Understand components of a VFD.
- Perform commissioning of a VFD.
- Create a SCADA control system.
- Actuator control using a SCADA system.
- Perform VFD control using a PLC and HMI.

Contents






Section A: PLC

- Introduction to PLC.
- Basic PLC wiring.
- Introduction to sensors and actuators.
- PLC programming overview.
- PLC troubleshooting and maintenance.

Section B: SCADA

- An Example of a SCADA System.
- Data Acquisition.
- SCADA system tags and graphic creation.
- Human Machine Interface (HMI).

Contents



Section C: VFD

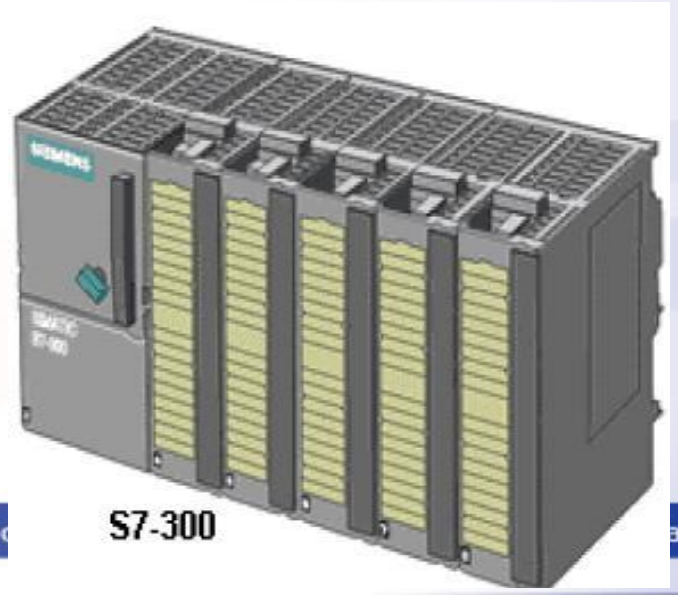
- Introduction to motor drive system design
- Overview of drive commissioning.
- Different methods of drive control.
- Drive control using a PLC in a SCADA system.



Section A:

Programmable Logic Controllers

A PLC (Programmable Logic Controllers) is an industrial computer used to monitor inputs, and depending upon their state make decisions based on its program or logic, to control (turn on/off) its outputs to automate
a machine or a process.





COMPONENTS OF PLC

A. Central Processing Unit

B. Memory unit

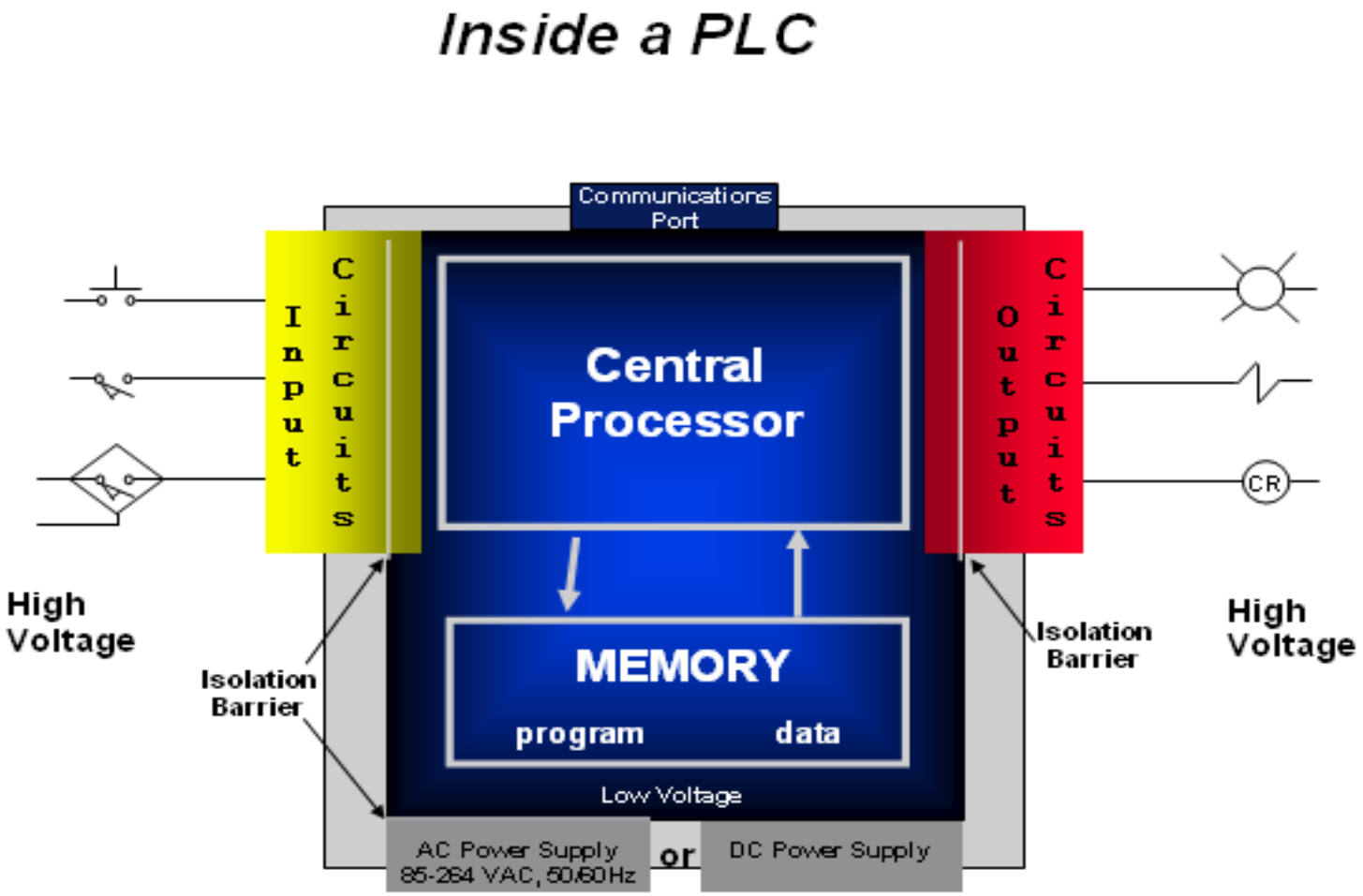
C. Power supply.

D. Input/output Modules.

E. Programming device.

Central Processing Unit





Processor:



A. Processor:

- Processor is the central processing units (CPU) of the programmable controller, it executes the various logic and sequencing functions by operating on the PLC inputs to determine the appropriate output signals.
- Microprocessor based, may allow arithmetic & logic operators, block memory moves, computer interface, local area network, functions, etc.
- CPU makes a great number of check-ups of the PLC controller itself so eventual errors would be discovered early

Memory unit:



B. Memory unit:

- *System (ROM)* to give permanent storage for the operating system and the fixed data used by the CPU.
- *RAM* Microprocessor based, may allow arithmetic & logic operators, block memory moves, computer interface, local area network, functions, etc.
- *EPROM* for ROM's that can be programmed and then the program made permanent.



Power supply:

C. Power supply:

- A power supply of 120v alternating current (ac) used to drive the PLC (some units operate on 240v ac)
- The power supply converts the 120v (ac) into direct current (dc) voltages of ± 24 v.
- These low voltages are used to operate equipment that may have much higher voltage and power ratings than the PLC itself.



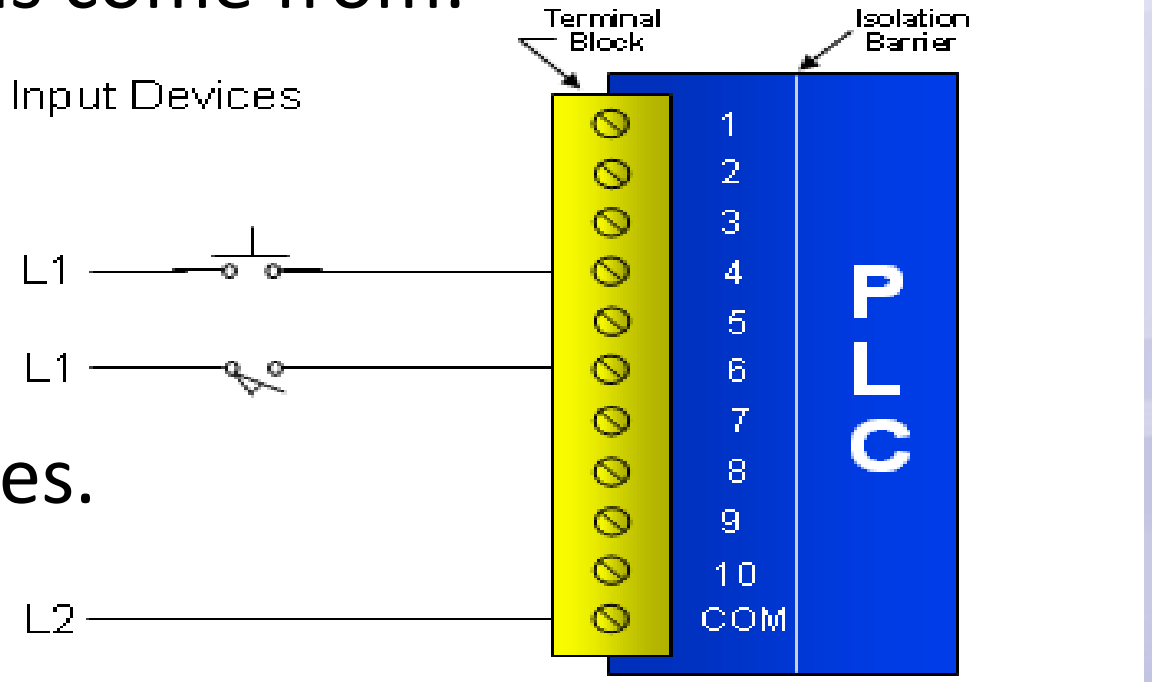
Signal Modules:

D. Input/Output Modules:

1) Input Module:

Input to the PLC are signals come from:

- Limit-switches,
- Pushbuttons,
- Sensors,
- And other on/off devices.








Signal Modules: Input Wiring

Sinking/Sourcing

Sinking sensors allow current to flow into the sensor to the voltage common, while sourcing sensors allow current to flow out of the sensor from a positive source. For both of these methods the emphasis is on current flow, not voltage. By using current flow, instead of voltage, many of the electrical noise problems are reduced.

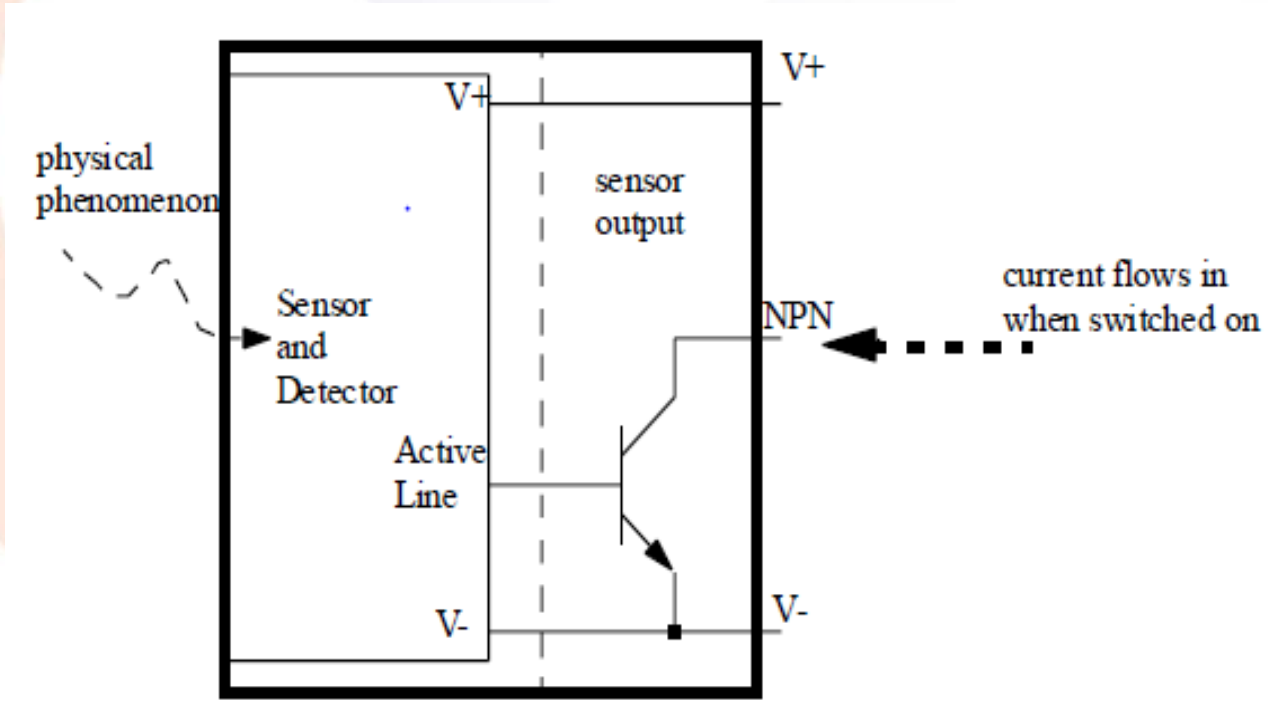


Sinking/Sourcing

When discussing sourcing and sinking we are referring to the *output* of the sensor that is acting like a switch. In fact the output of the sensor is normally a transistor, that will act like a switch (with some voltage loss). A PNP transistor is used for the sourcing output, and an NPN transistor is used for the sinking input.

Sinking

When discussing these sensors the term sourcing is often interchanged with PNP, and sinking with NPN. A simplified example of a sinking output sensor is shown in Figure





Sinking

The sensor needs a voltage supply to operate, so a voltage supply is needed for the sensor. If the sensor has detected some phenomenon then it will trigger the active line. The active line is directly connected to an NPN transistor.

(Note: for an NPN transistor the arrow always points away from the center.) If the voltage to the transistor on the *active line* is 0V, then the transistor will not allow current to flow into the sensor. If the voltage on the active line becomes larger (say 12V) then the transistor will switch on and allow current to flow into the sensor to the common.

Sourcing





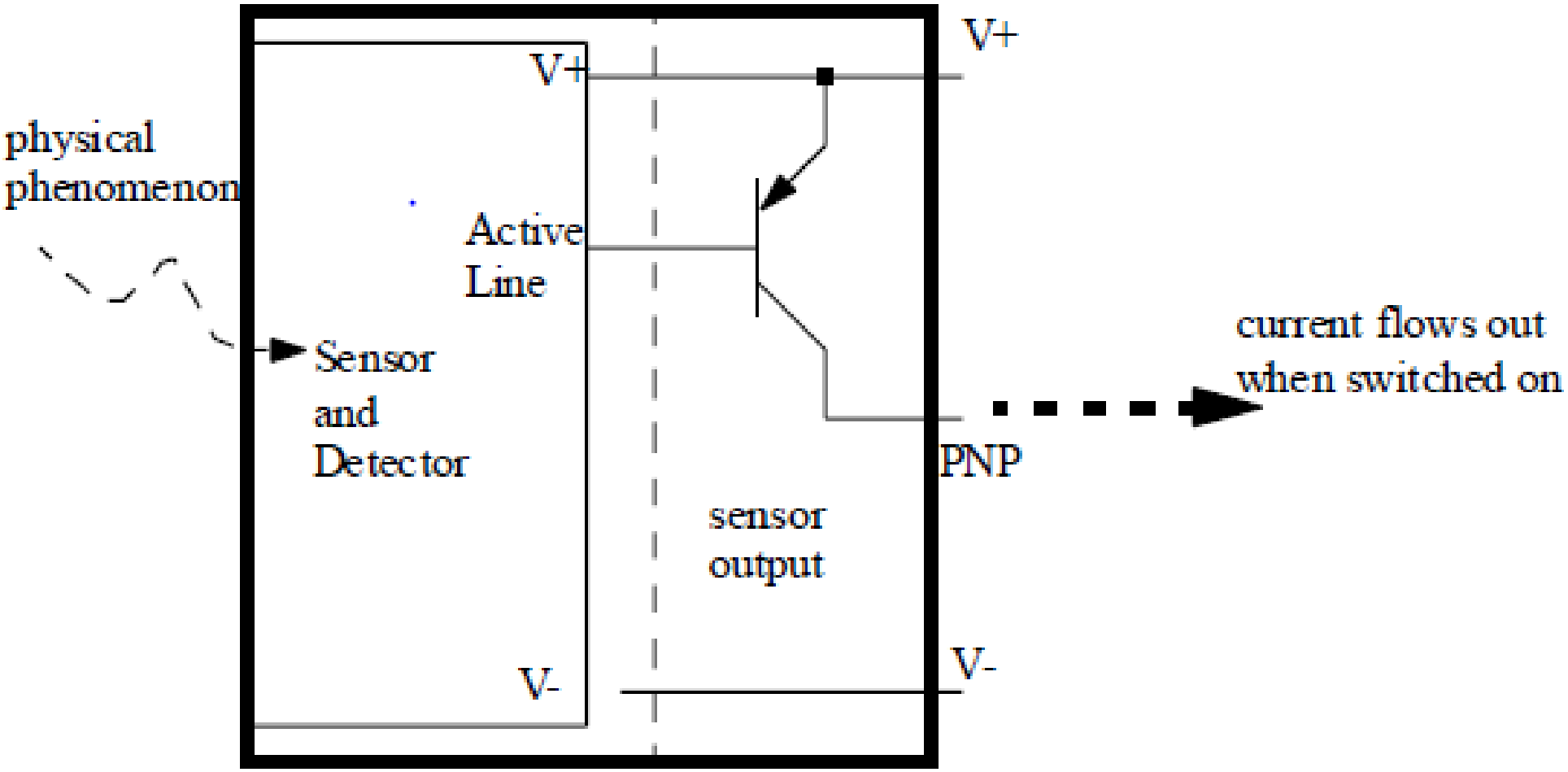




The sourcing sensors use a PNP transistor, (Note: PNP transistors are always drawn with the arrow pointing to the center.) When the sensor is inactive the active line stays at the V+ value, and the transistor stays switched off. When the sensor becomes active the active line will be made 0V, and the transistor will allow current to flow out of the sensor.

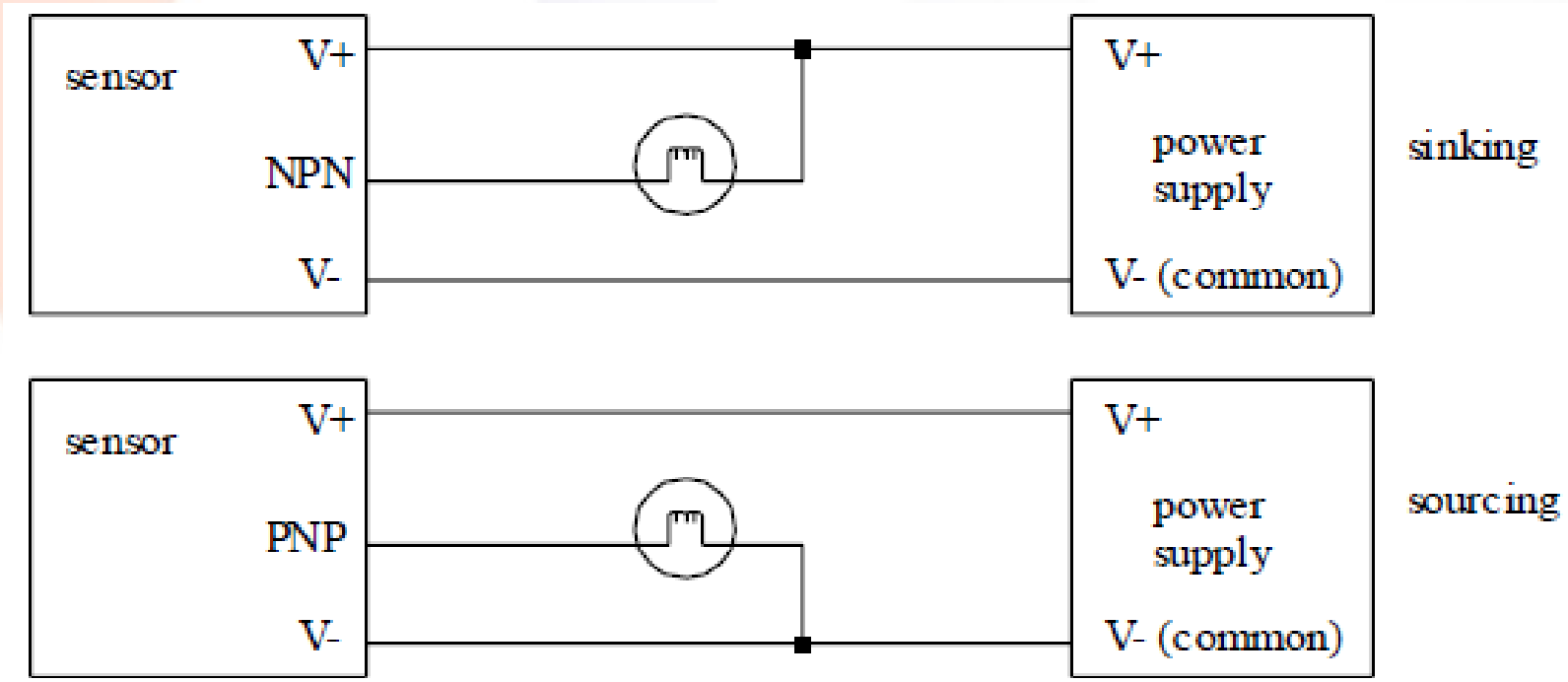


Sourcing

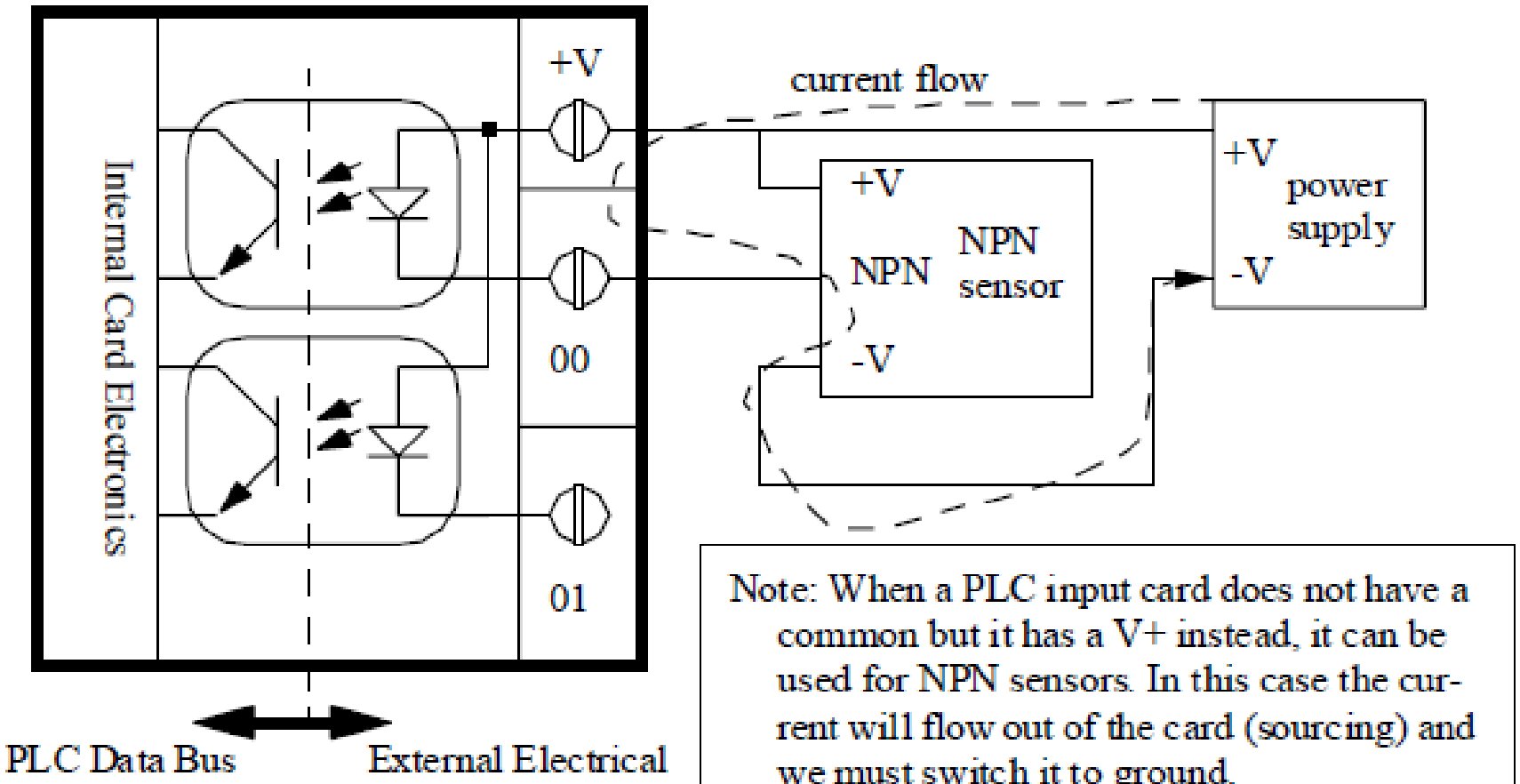

V+
V+
physical
phenomenon
Active
Line
Sensor
and
Detector
current flows out
when switched on
PNP
sensor
output
V-
V-

Sensor test for PNP and NPN

Most NPN/PNP sensors are capable of handling currents up to a few amps, and they can be used to switch loads directly

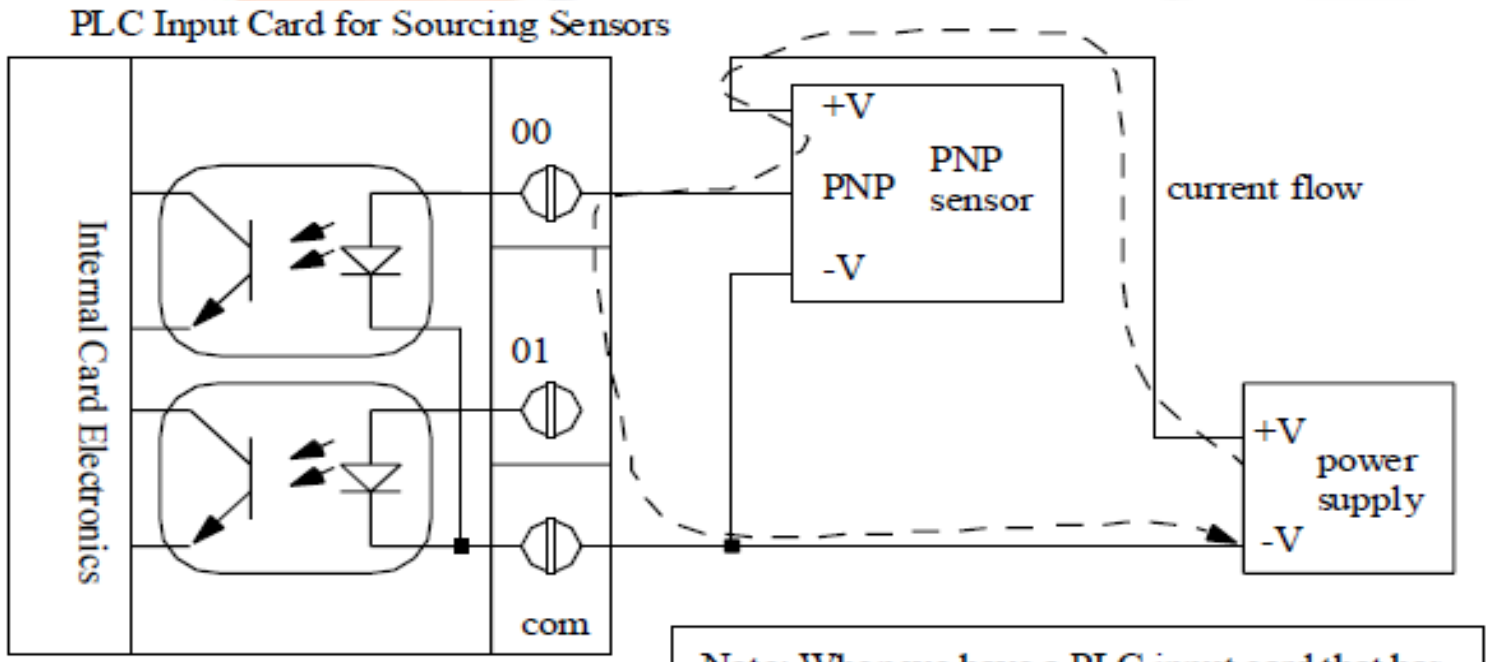


PLC Input Card for Sinking Sensors



Note: When a PLC input card does not have a common but it has a V+ instead, it can be used for NPN sensors. In this case the current will flow out of the card (sourcing) and we must switch it to ground.

PLC Input Card for Sourcing Sensors



Note: When we have a PLC input card that has a common then we can use PNP sensors. In this case the current will flow into the card and then out the common to the power supply.

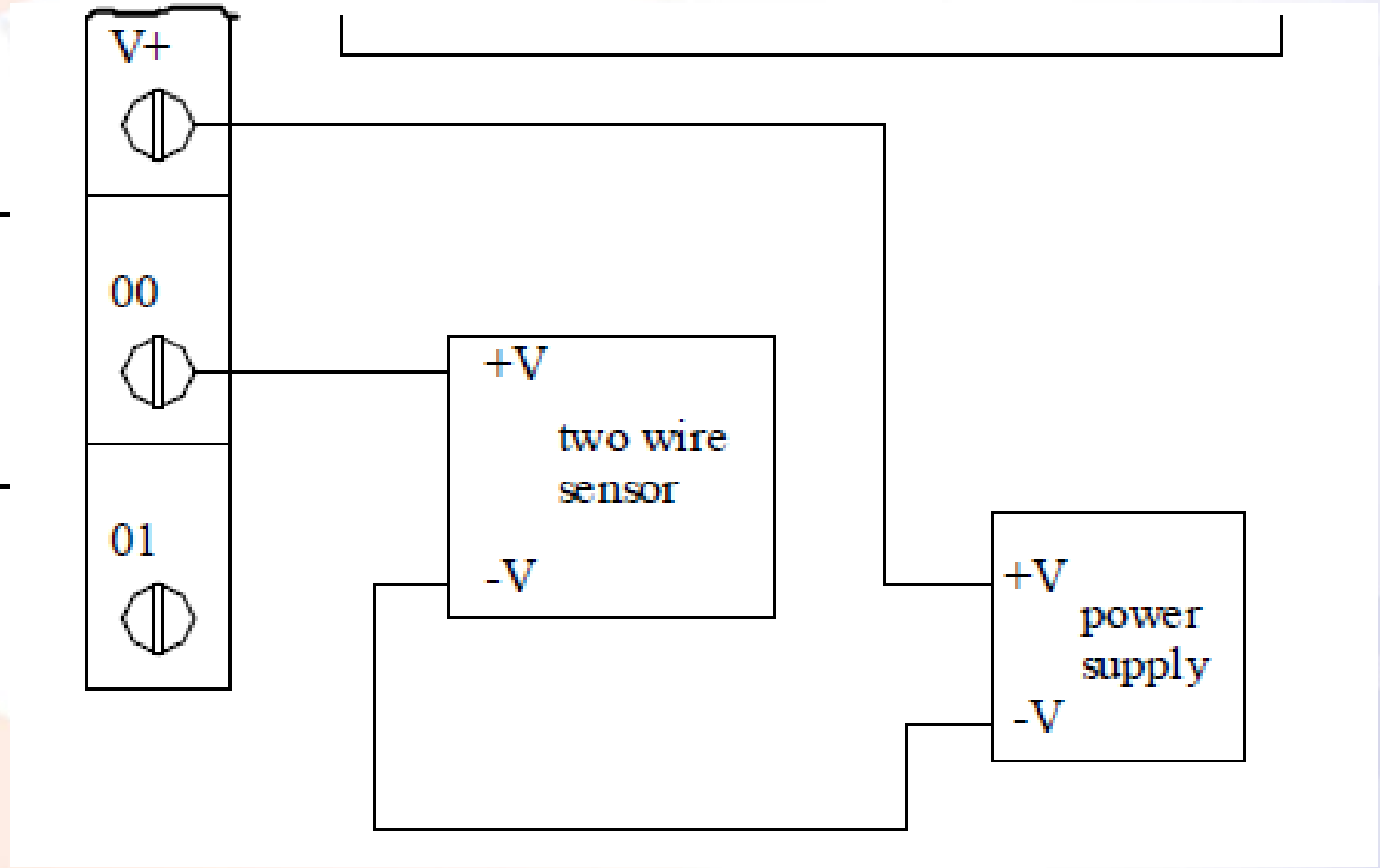


Two wire sensors

A two wire sensor can be used as either a sourcing or sinking input. In both of these arrangements the sensor will require a small amount of current to power the sensor, but when active it will allow more current to flow. This requires input cards that will allow a small amount of current to flow (called the leakage current), but also be able to detect when the current has exceeded a given value.

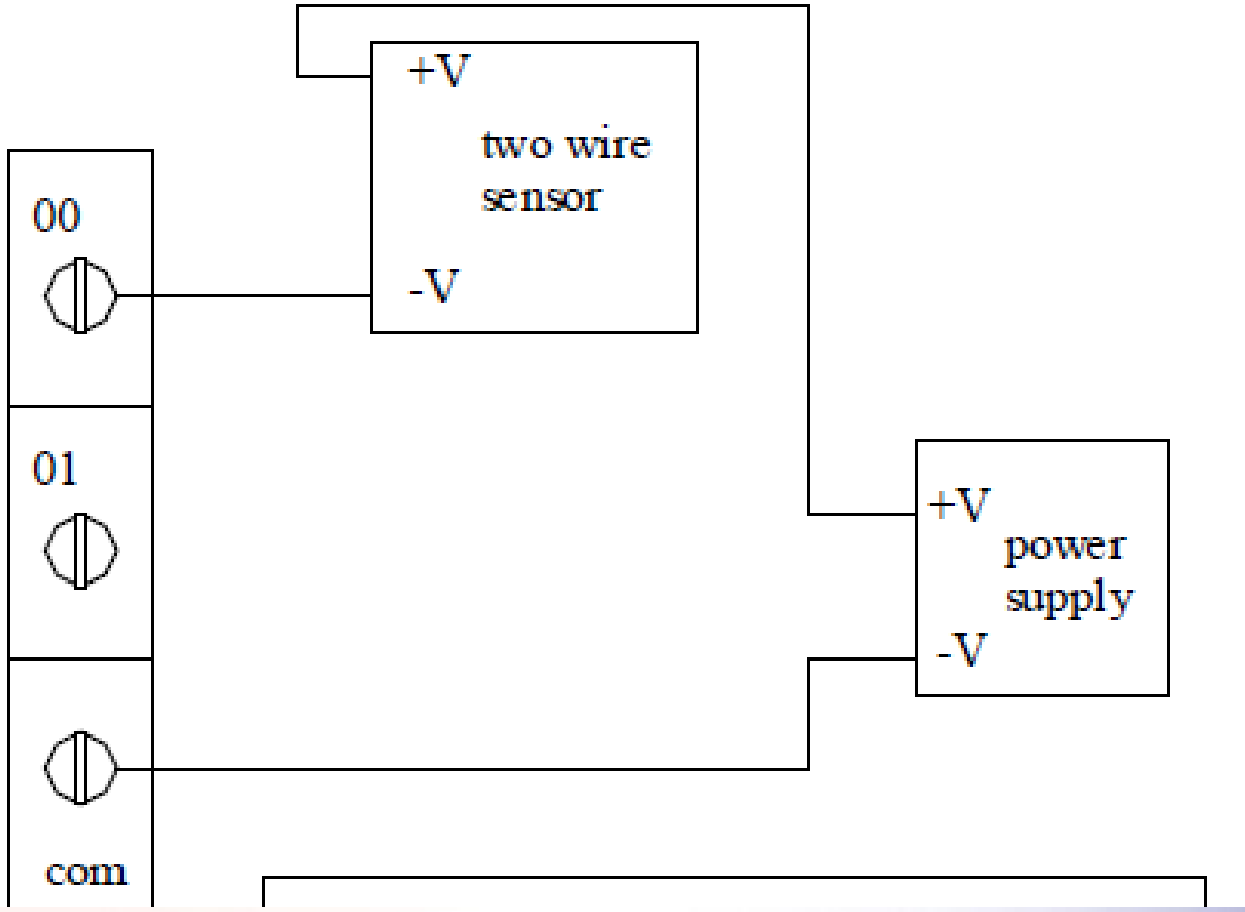
PLC Input Card for Sinking Sensors

Note: These sensors require a certain leakage current to power the electronics.



PLC Input Card for Sourcing Sensors

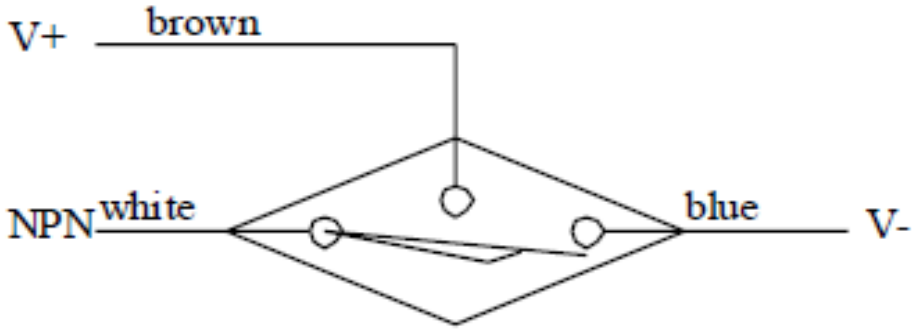
Note: These sensors require a certain leakage current to power the electronics.



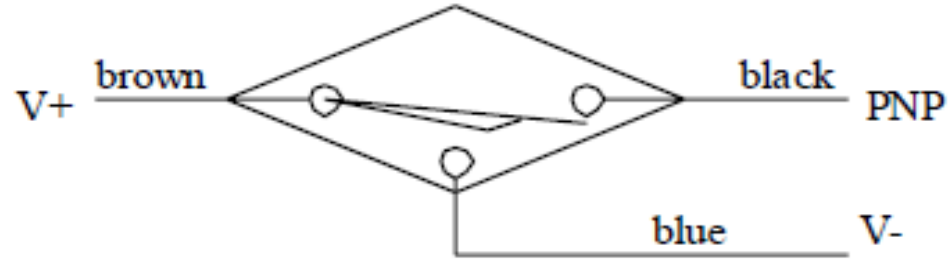
Sourcing and Sinking Schematic Symbols



NPN (sinking)



PNP (sourcing)

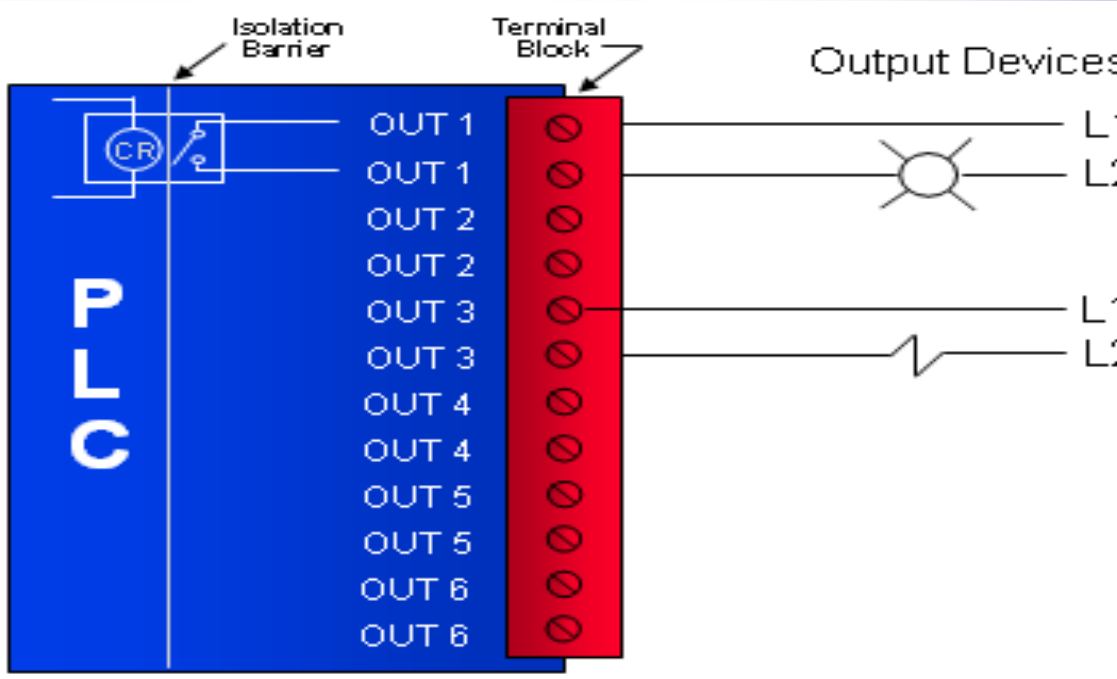




Signal Modules:

2) Output Module:

- Outputs from the controller are on/off signals to operate:
  - Motors,
  - Valves,
  - And other devices required to actuate the process.



DC and Relay Output Wiring

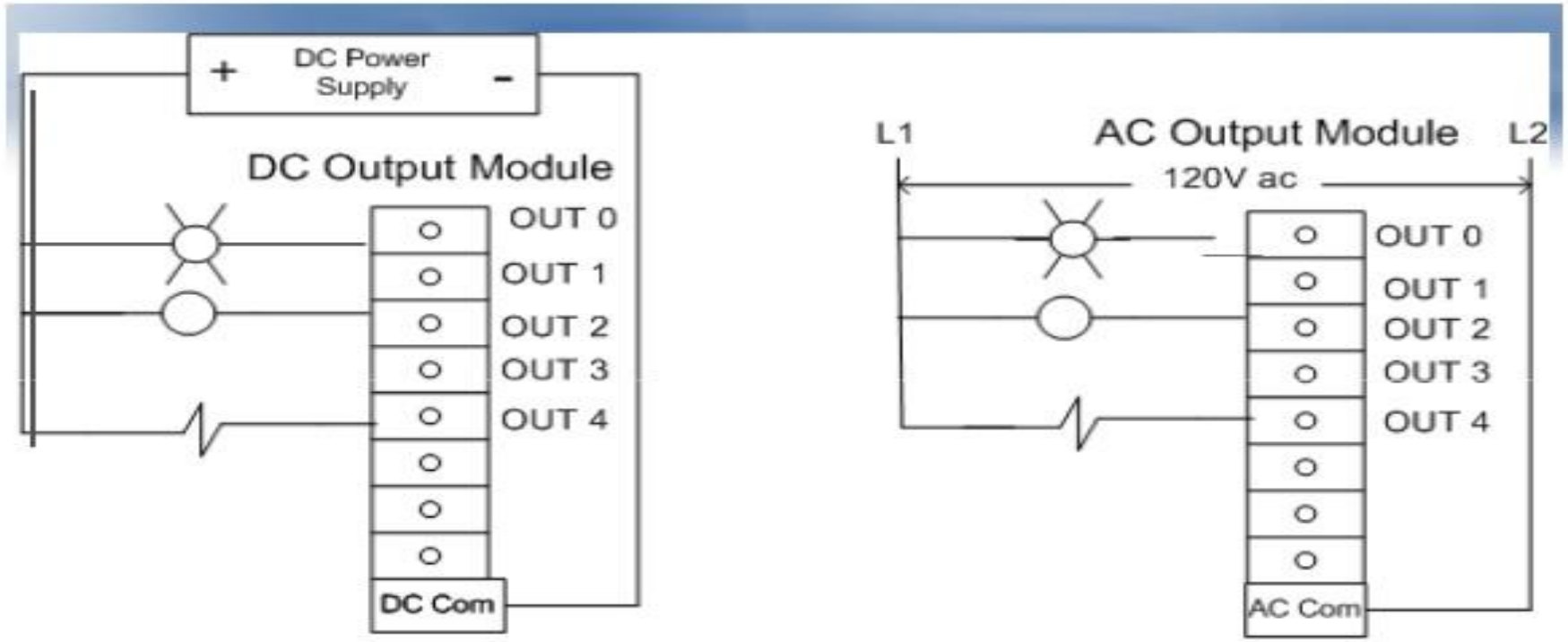
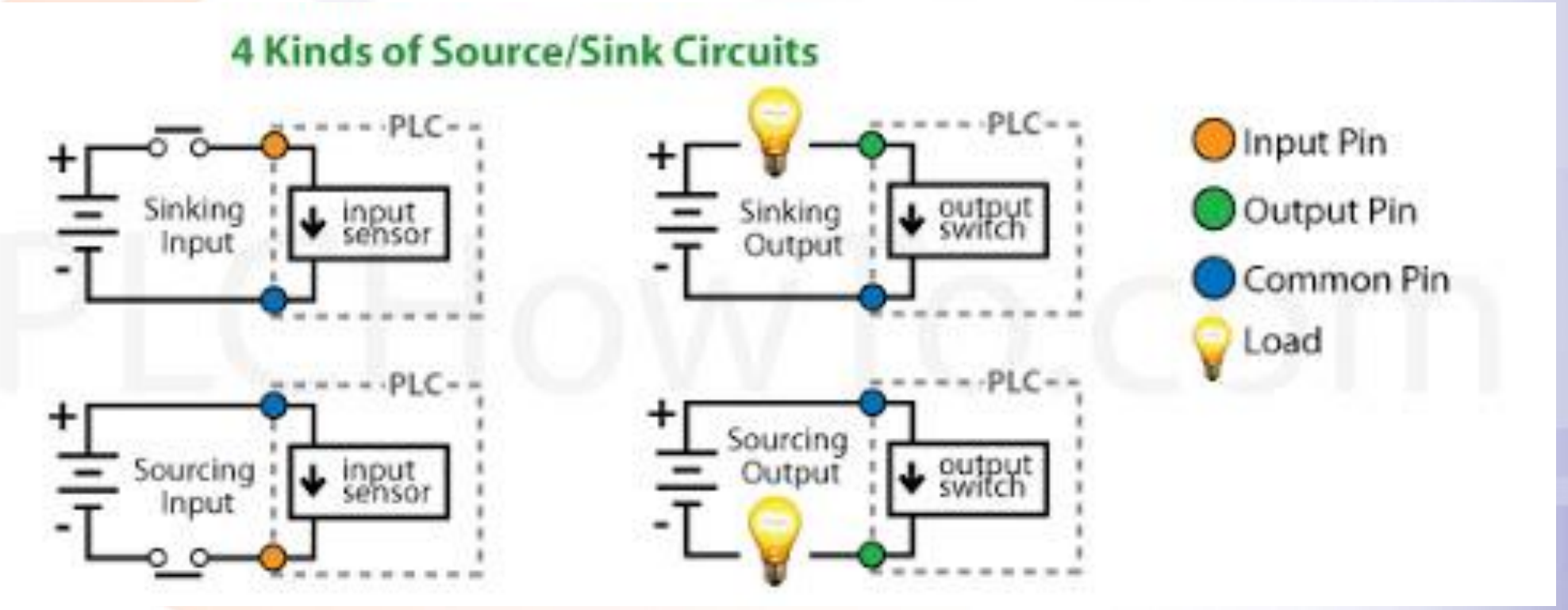


Figure 3.8 DC OUTPUT WIRING CONNECTIONS

Figure 3.9 AC OUTPUT WIRING CONNECTIONS



Sinking and Sourcing I/O Wiring

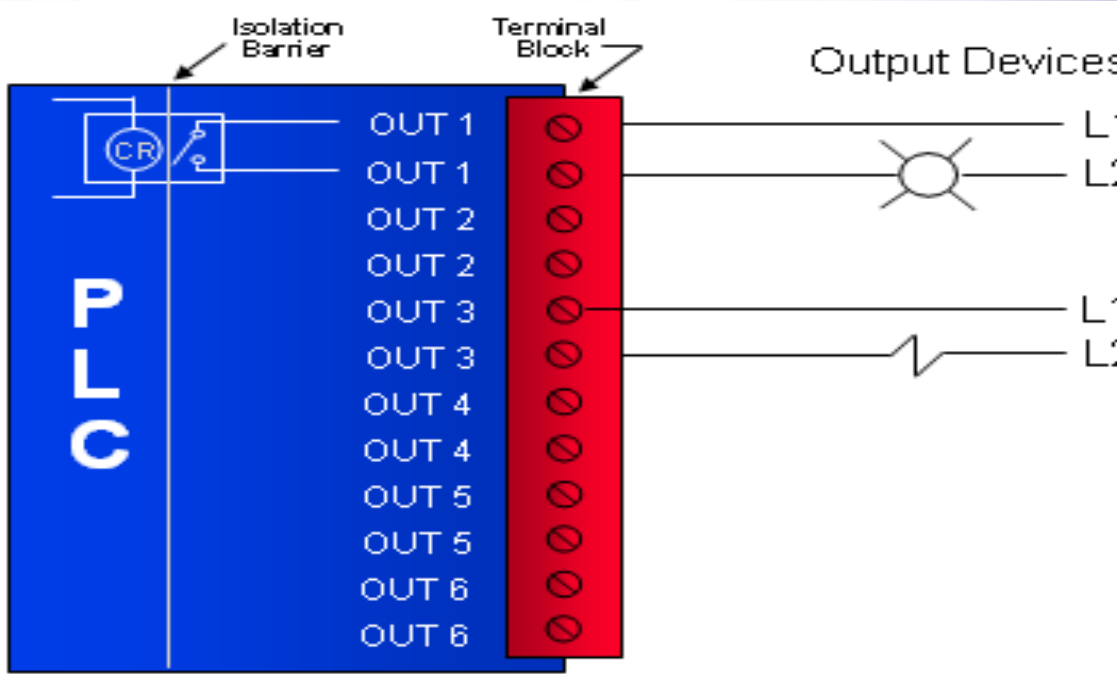




Signal Modules:

2) Output Module:

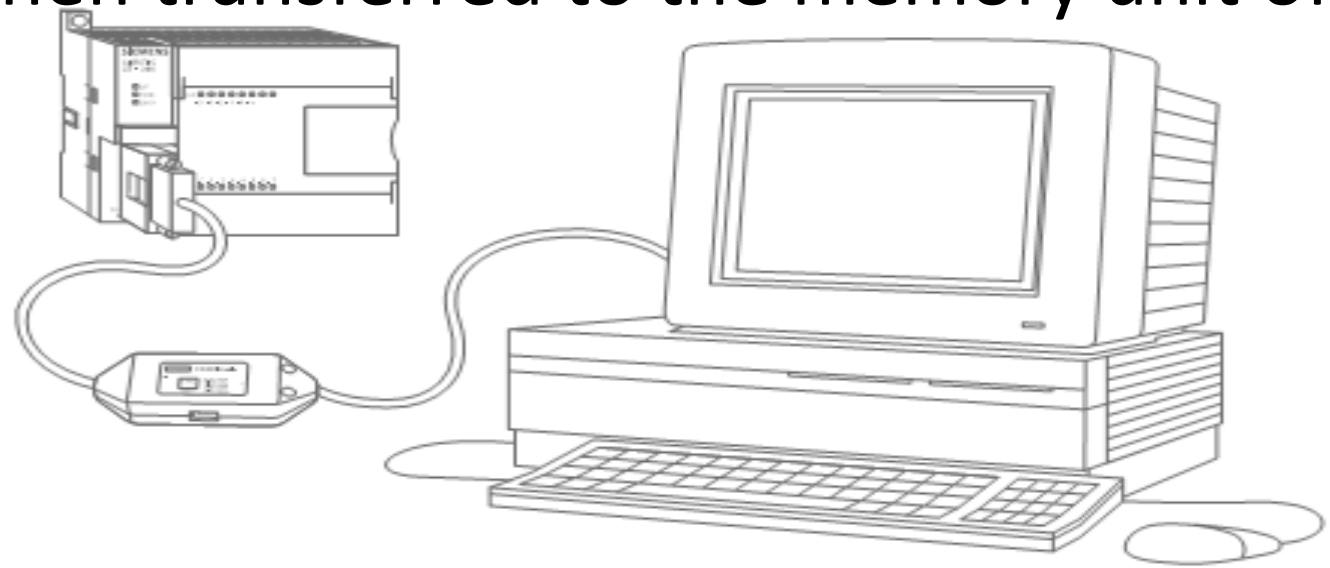
- Outputs from the controller are on/off signals to operate:
  - Motors,
  - Valves,
  - And other devices required to actuate the process.



Programming device:

E. Programming device:

- Programming device is used to entered the required program into the memory of the processor.
- The program is developed in the programming device and then transferred to the memory unit of the PLC.



Types of PLC



- Compact PLC
  - Fixed casing
  - Fixed number of I/O (most of them binary)
  - Limited process computer capabilities
- Modular PLC
  - Modular construction (backplane)
  - One- or multiprocessor system
  - Large variety of input/output boards

Compact PLCs





Allen Bradley



Siemens

Omron



Modular PLCs



Allen Bradley



PLC Operations

The PLC program is executed as part of a repetitive process referred to as a SCAN CYCLE.





PLC SCAN CYCLE

A PLC scan starts with the CPU reading the status of inputs. The application program is executed using the status of the inputs. Once the program is completed, the CPU performs internal diagnostics and communication tasks. The scan cycle ends by updating the outputs, then starts over. The cycle time depends on the size of the program, the number of I/Os, and the amount of communication required.

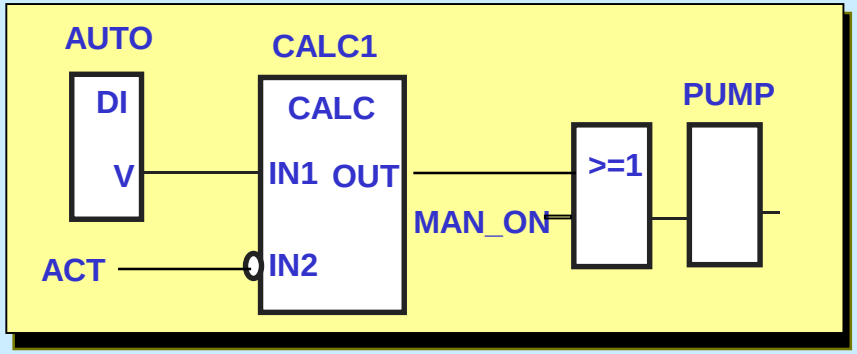
Typical SCAN CYCLE time is 100-150ms

The five IEC 61131-3 Programming languages

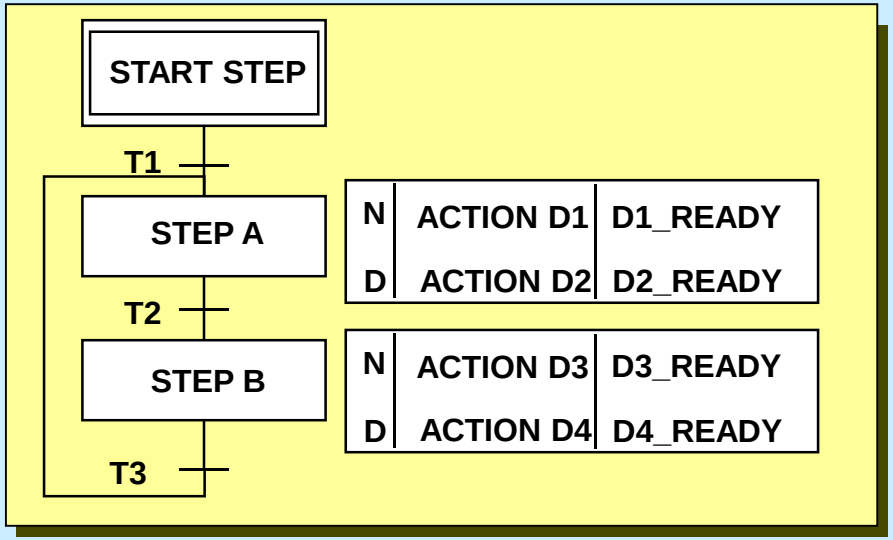


graphical languages

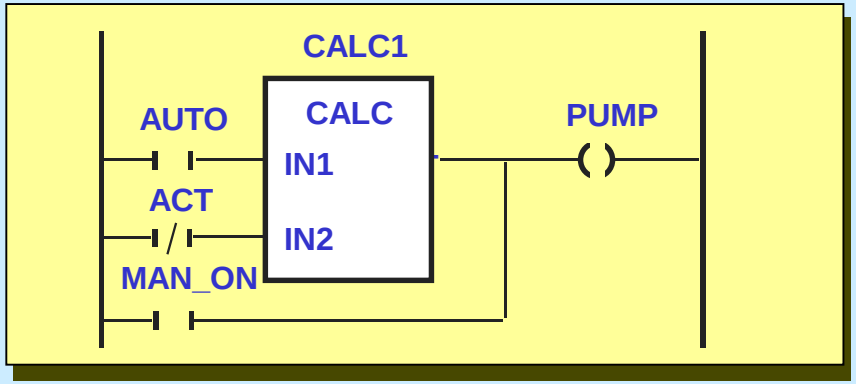
Function Block Diagram (FBD)



Sequential Flow Chart (SFC)



Ladder Diagram (LD)



textual languages

Structured Text (ST)

```
VAR CONSTANT X : REAL := 53.8 ;
Z : REAL; END_VAR
VAR aFB, bFB :  FB_type; END_VAR

bFB(A:=1, B:='OK');
Z := X - INT_TO_REAL (bFB.OUT1);
IF Z>57.0 THEN aFB(A:=0, B:="ERR");
ELSE  aFB(A:=1, B:="Z is OK");
END_IF
```

Instruction List (IL)

```
A: LD    %IX1  (* PUSH BUTTON *)
   ANDN  %MX5  (* NOT INHIBITED *)
   ST    %QX2  (* FAN ON *)
```






Section B:

SUPERVISORY CONTROL AND DATA ACQUISITION

- An Example of a SCADA System.
- Data Acquisition.
- SCADA system tags and graphic creation.
- Human Machine Interface (HMI).






Section A:

What is SCADA?

SCADA

is Supervisory Control And Data Acquisition”

- Real time industrial process control system used to centrally monitor and control remote or local industrial equipment such as motors, valves, pumps, relays, etc



TYPES OF SCADA SYSTEMS

- Adroit
- National Instruments : Labview
- Siemens : WIN CC
- Allen Bradly : RS View
- Modicon (Telemecanique) : Visual look
- Wonderware : Intouch






BASIC FUNCTIONS

- Data Acquisition
- Data Communication
- Data Presentation
- Control



More functions

- Alarming -On screen and SMS alarm facility for high level alarms
- History- trip status of equipment
- Reporting on quality, maintenance, production
- User friendliness
- Maximum demand monitoring



Why the need for monitoring

- Monitoring of process values assists a company in detecting of errors occurrence in order to avoid system deviation hence causing production downtime.
- Monitoring also provides system backup through data logging of process values.



Devices used in monitoring

- In order to monitor process values we use devices known as Front to End Devices FEDs and Remote terminal Units RTUs such as PLC, ADAMS
- Monitoring is all about communication therefore there is a need to configure some of these communication modules



Economic value

- Energy efficiency through logging energy consumption and generating trends
- Performance optimization through real time data evaluation
- Downtime reduction through timely fault diagnostics



The need for SCADA

- Do you need real time status of devices in a large complex system?
- Is there equipment in your network that you want to monitor from a distance?
- Do you need to measure how changing inputs affect the change in your operations?

APPLICATION OF SCADA SYSTEMS









SCADA is used in controlling:

- Natural gas distribution
- Electric generation/distribution
- Water storage, purification and distribution
- Lighting
- Heating, cooling and ventilation
- Fire alarm and suppression
- Elevators and escalators
- Process line control



SCADA COMPONENTS

- HMI/ Monitor for display
- Supervisory system(computer)
- Field End Devices/ Data Acquisition Devices
- Communication Interface

SCADA Hardware



- Field level instrumentation and control devices

  Sensing field variables (sensors), controlling processes (actuators)

- Remote Terminal Units (RTUs)

  Send control and monitoring data to MTU (could be a PLC)

- Master Terminal Unit (Also called Master Station)

  Gather information from RTUs, monitoring and processing, sending necessary control actions to RTUs



SCADA Hardware

- Communication system

  The communication between the various components, MTU, RTU can be wireless or wired, wireless-wifi, radio, wired-RS485/RS422/RS232, Ethernet



Communication Interface

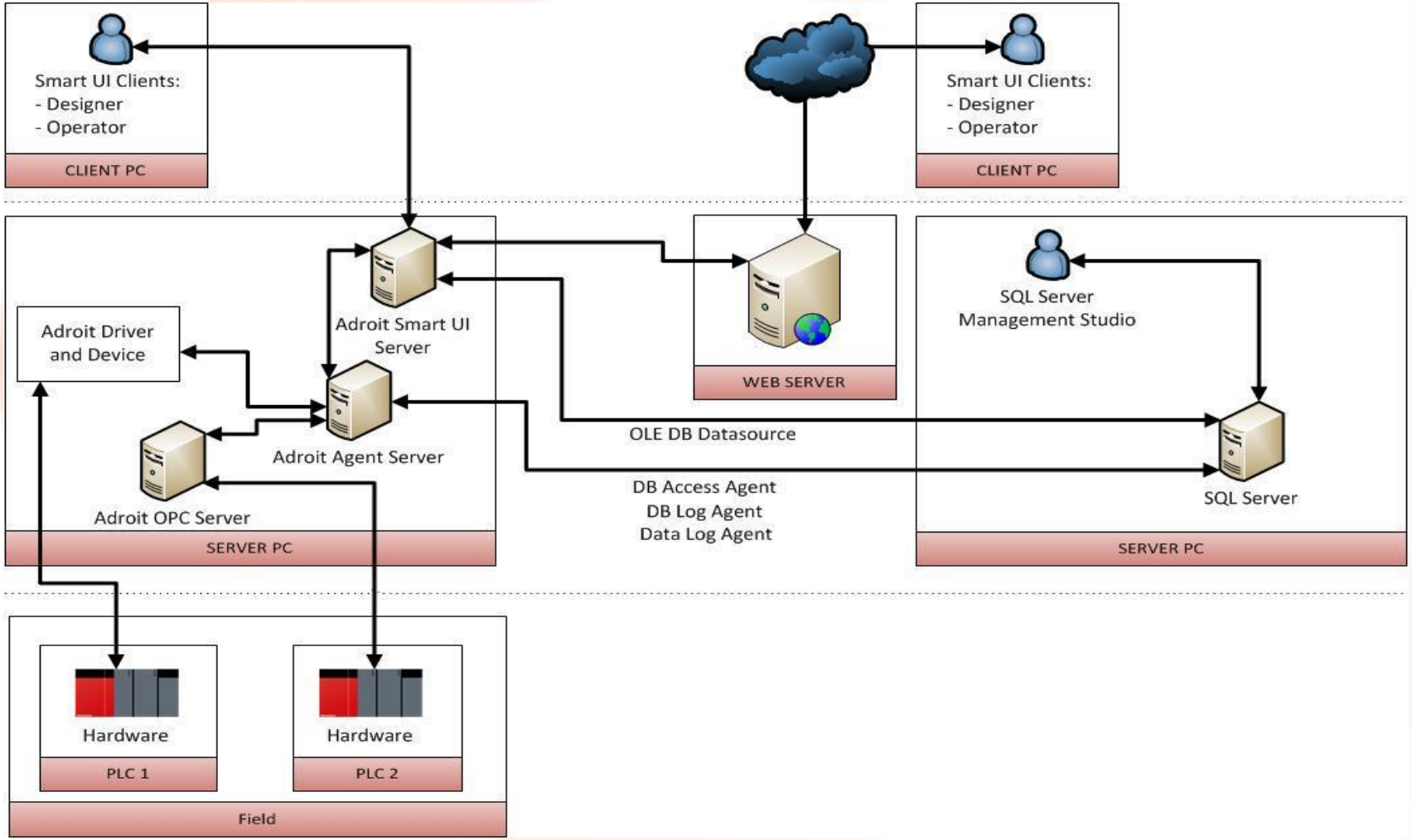
- Serial cables (RS 232, RS 485, RS 422)
- USB
- Ethernet cable
- Radio comm.
- Optical cable
- MPI point to point



Requirements for Scada

- Clear data presentation
- Programmable control
- Multiprotocol support
- Nuisance alarm filtering

SCADA ARCHITECTURE













Adroit

Smart SCADA



ADROIT

A

Distributed

Real-time

Object-Oriented

Intelligent

Toolkit



ADROIT

- Adroit is an open , object oriented , process automation toolkit, structured a distributed client-server architecture where:

a) The server component is the Agent Server

b) The client component is typically the Smart UI



What can be done with Adroit?

- Adroit is a SCADA product and as such is primarily concerned with process values since these are essentially used to supervise and control processes

Adroit provides the following value operations






- Agent Architecture

  Creates the necessary Agents to both acquire and store the process variables and perform more advanced calculations and manipulations.

- Communication of process values

  This is primary function of any SCADA, in that it allows a user to bring in and send out values via front end devices in the field. This allows the process to be supervised (SCADA).

Adroit provides the following value operations



- Alarming of values

  when values transgress configurable operational limits, an alarm can be created to notify the user of this transgression using a variety of methods.

- Logging of recording values

  This allows the values of the process to be saved to proprietary historic files and or databases so that they can be retrieved for long term analysis and reporting purposes real time trending.

Adroit provides the following value operations



- Displaying of values

  This is usually performed through the creation of graphic forms where values can be displayed, manipulated or used to animate the GF's.



AGENT

- An agent is the basis and the building block of Adroit residing within the AS.
- Agents have intelligence in that they know how to manipulate data.

Examples of agents

Analog, Digital, Marshall, Expression, Alarm



Adroit Programming Windows

Agent Server

- The AS is the equivalent of an I/O server in SCADA systems and performs the real-time data acquisition ,storage and alarming of all data from devices in the field



Adroit Programming Windows

Configurator

- Is the central point from which to configure Agents in a local or remote AS from Adroit

Smart UI Designer

- The working space for the project
- It contains graphics and allows for behaviors to be assigned to them



Adroit Programming Windows

Smart UI Operator

- The monitoring area once the design is complete

Smart UI Server Configuration

- Used to configure default settings for the designer and operator.



Client Server Architecture

- Provides the basis of building large distributed systems. Multiple processors connected via LAN network can be distributed throughout a plant, sharing info in a typical client server architecture. Messages,files,commands and status info can easily be transferred between computers.

SCANNING






- This is the sending of or communication of values between Adroit and the FEDs in the field such as PLC and RTUs.
- FEDs use different protocols or languages for communication, Adroit has developed a large no of protocols drivers which allow for communicate.
- Scanning is managed in Adroit by means of active specialist agents called scan Agents. A total of 65334 device agents may be configured from a single driver for communication to the FED



Devices and Scanning Setup

- Devices i.e. PLCs, Protocol OPC
- Communication to PLCS MP,ETH,SL,OPC
- Contiguous data areas-every effort must be made when drawing the I/O schedule of the PLC,SCADA to define logical and contiguous data areas to be scanned from devices






Section B:

Variable Frequency Drives

- Introduction to motor drive system design
- Overview of drive commissioning.
- Different methods of drive control.
- Drive control using a PLC in a SCADA system.



What is VFD?

This technology has several different names such as :

- Variable speed drives.
- Adjustable speed drives.
- Adjustable frequency drives.
- Frequency converter.

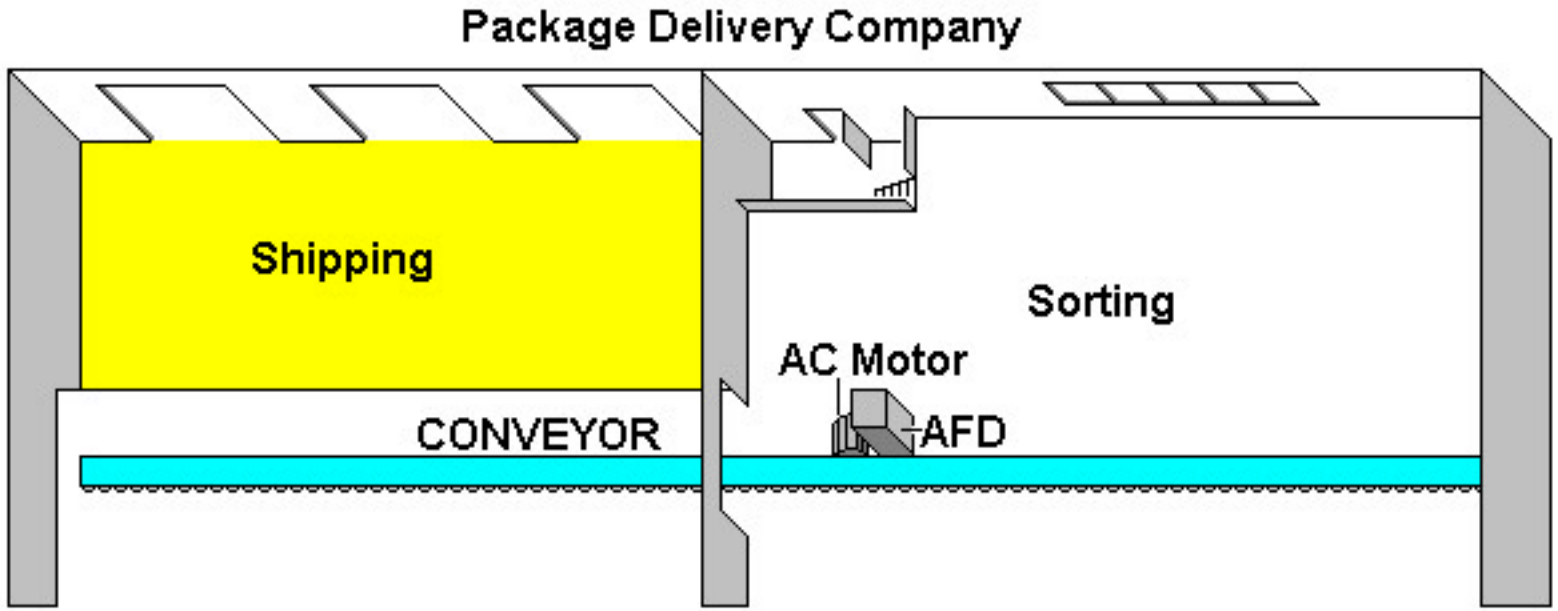


Introduction

- AC Adjustable Frequency Drives can be thought of as electrical control devices that change the operating speed of a motor.
- VFD’s are able to vary the operating speed of the motor by changing the electrical frequency input to the motor.

Introduction

To understand the functions of an AFD better, an example of a conveyor at a Package Delivery Company is used.





Basic VFD functions

A few of the basic functions of an AFD in controlling the AC motor and conveyor are covered here.

1. Bi-Directional Operation

- FORWARD

One function of the AFD is to operate the motor in a forward direction, to move the packages from the shipping area to the sorting area.



Basic VFD functions

- REVERSE

Power going to the motor must be changed to move the packages backward (Reverse) from the sorting area back to the shipping area. If there were no AFD, 2 of the 3 leads of the 3-phase motor would be switched in order for the motor to change its direction and go backwards.



Basic VFD functions

2. Change Speed

The speed of the conveyor must be adjustable to allow for a slower speed when fewer packages and employees are present and a faster speed for large volumes. This allows the operator to match the speed of the conveyor to a particular process. The setting of this speed is known as the Reference.

Commonly reference refers to speed in Hertz (Hz), maximum reference of 60Hz, and minimum reference of 0Hz.



Basic VFD functions

3. Constant Speed

Another function of the AFD is to maintain the speed of the conveyor regardless of the number of packages. The AFD automatically compensates the current and torque to accommodate changes in the load.

Basic VFD functions

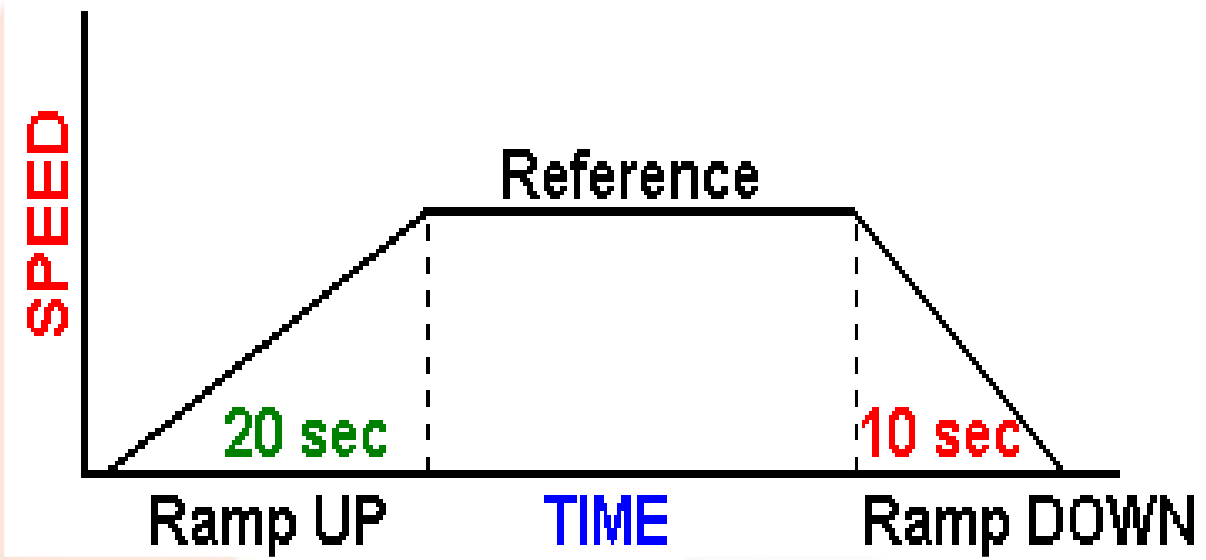


5. Ramping

The AFD also ramps the conveyor up and ramps it down. When the conveyor starts, acceleration, it is important that there is no sudden lurch to the reference speed, or packages can go rolling backward. Ramp-up slowly increases the speed from stopped or 0 Hz up to the reference, 34 Hz, over a certain amount of seconds perhaps 10 seconds. If this ramp is too short, the drive can trip on an over current alarm or torque limit.

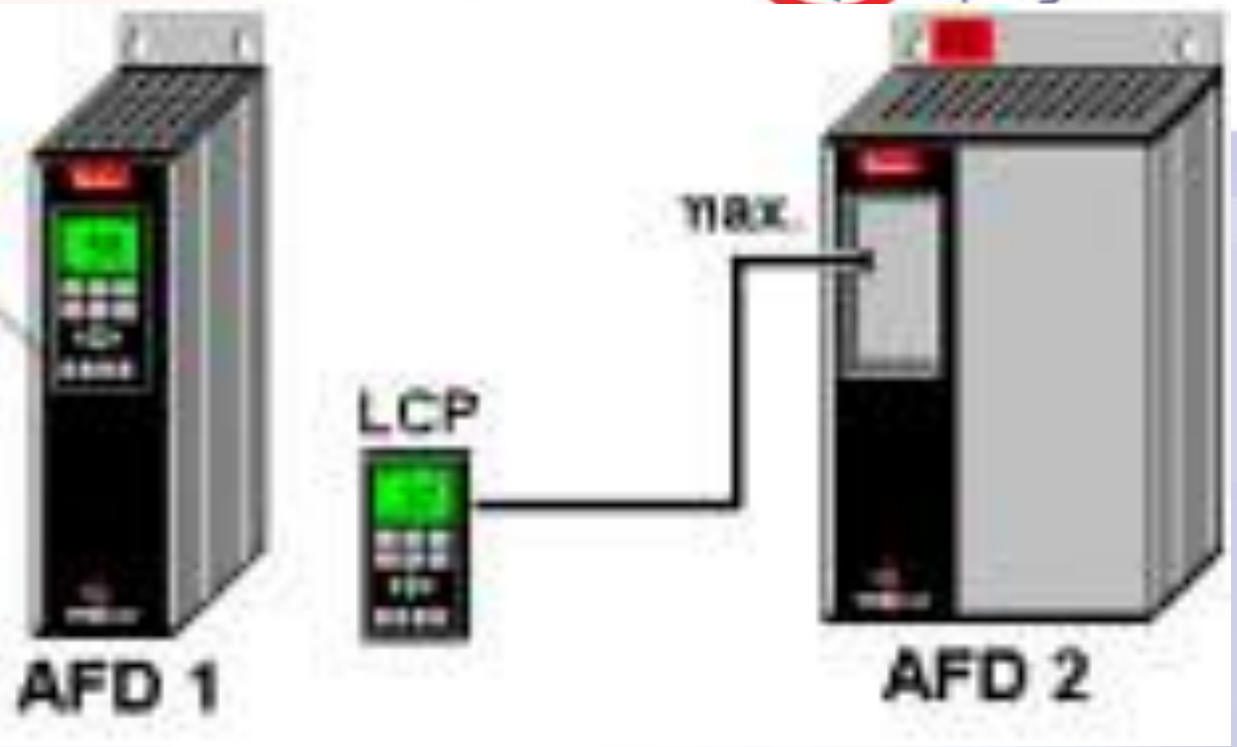


Basic VFD functions



Control of a VFD



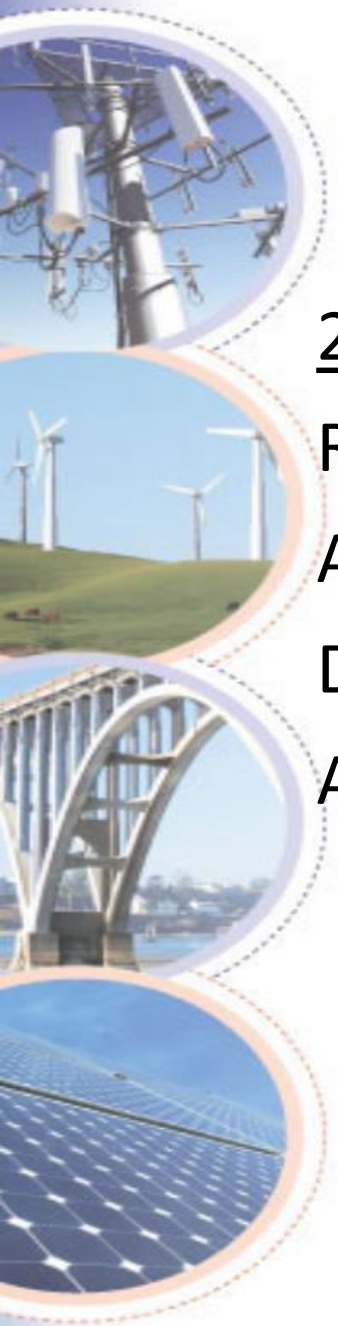


1) Local Control

Start key starts the drive

Stop/Reset key stops the drive and also resets the drives after an alarm. FWD/REV changes direction

Jog is a pushbutton that runs at a low frequency, say at 10Hz, usually used during setup.



Control of a VFD

2) Remote Control

Remote signals are wired into the control section of the AFD. They come in 4 types:

Digital Inputs, Analog Inputs,

Analog Outputs, Digital/Relay Outputs.

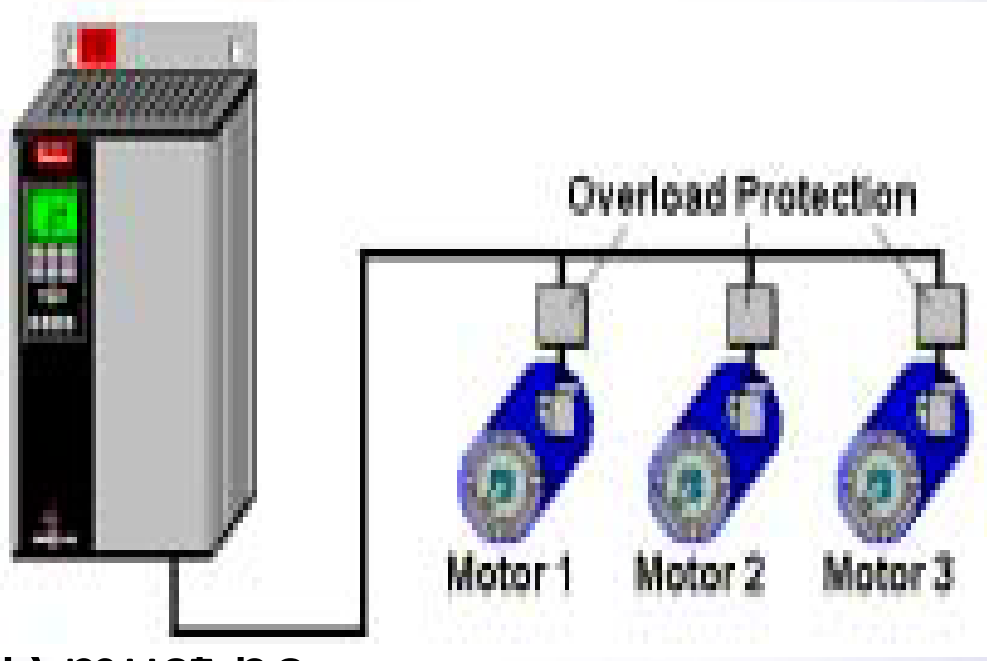


Control of a VFD

3) Multi-Motor Operation

One AFD is used to operate 3 separate AC motors. The AFD must be able to handle the maximum current for the 3 motors. Some features of the AFD in this arrangement are restricted.

Control of a VFD



In the example above each of the 3 AC motors is the same size and has a maximum current rating, Full Load Amps (FLA) of 5 amps. The AFD must be sized for 15 amps. Note individual overload protection on each motor.

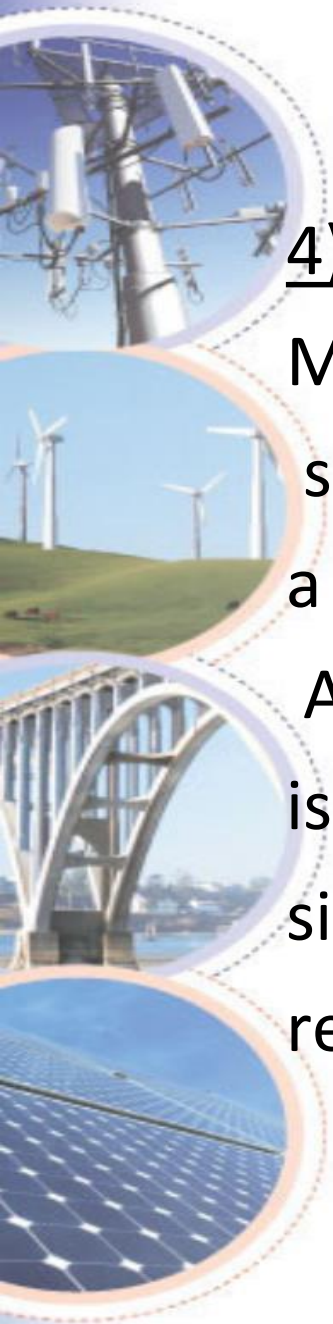


Control of a VFD

4) Master/Slave (Leader/Follower)

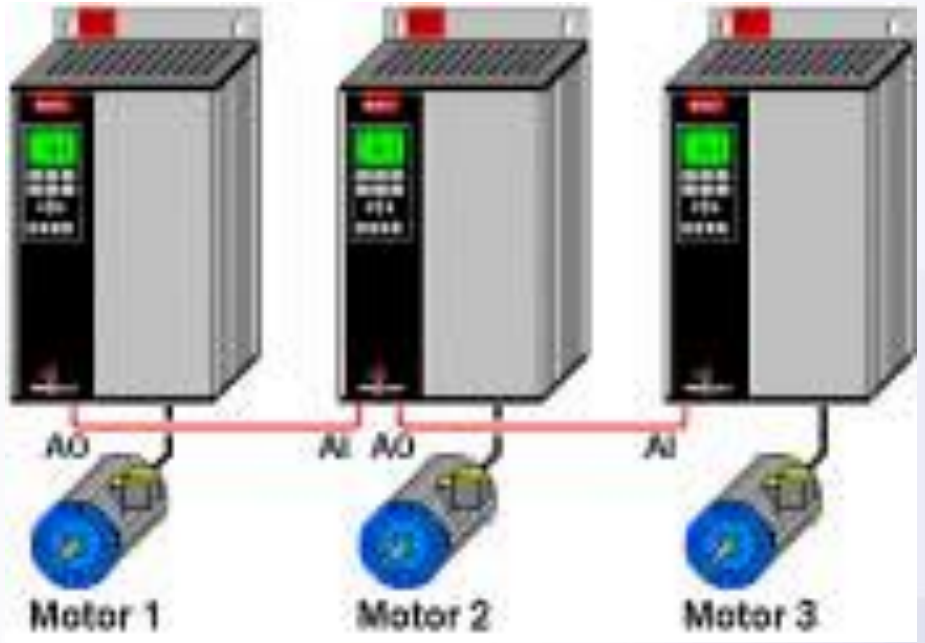
In the Master/Slave arrangement a Master or Lead AFD is used to operate a single motor and send its corresponding signal to the other 2 following motors. Features of the AFD which were restricted in the multi-motor can be used in this arrangement.

Control of a VFD



4) Master/Slave (Leader/Follower)

Motor 1, the Master/Leader, sends a reference using a 4-20mA signal from its AO to the AI of motor 2. It is also possible to send a pulsed signal as a reference.

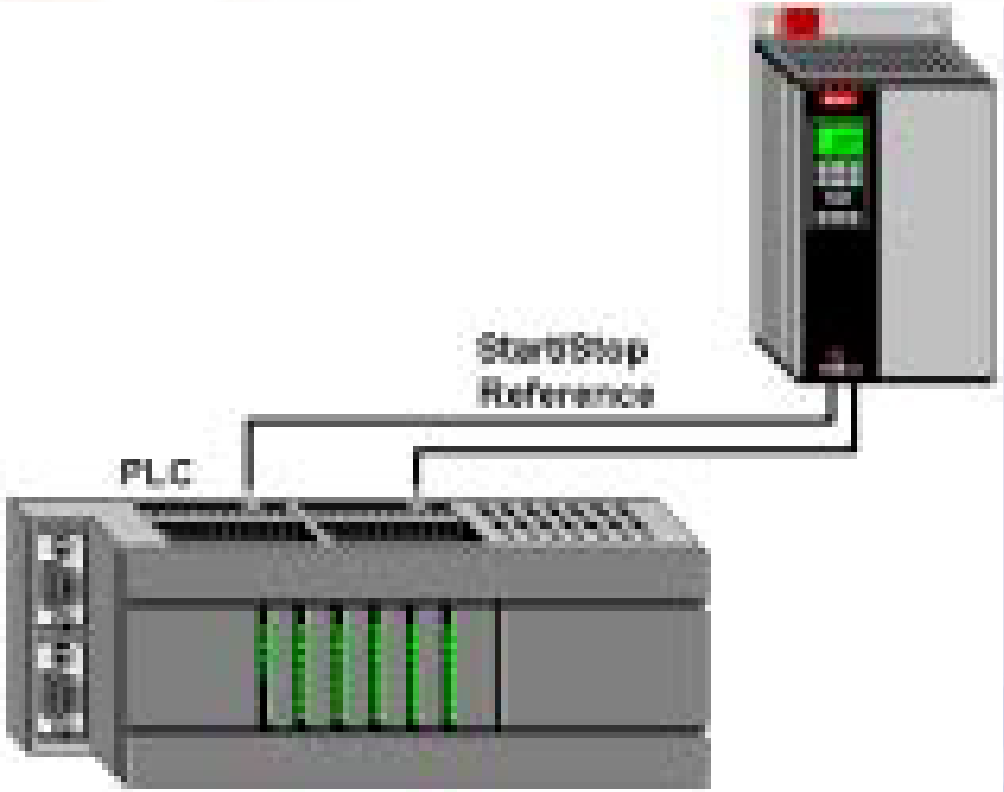


Control of a VFD



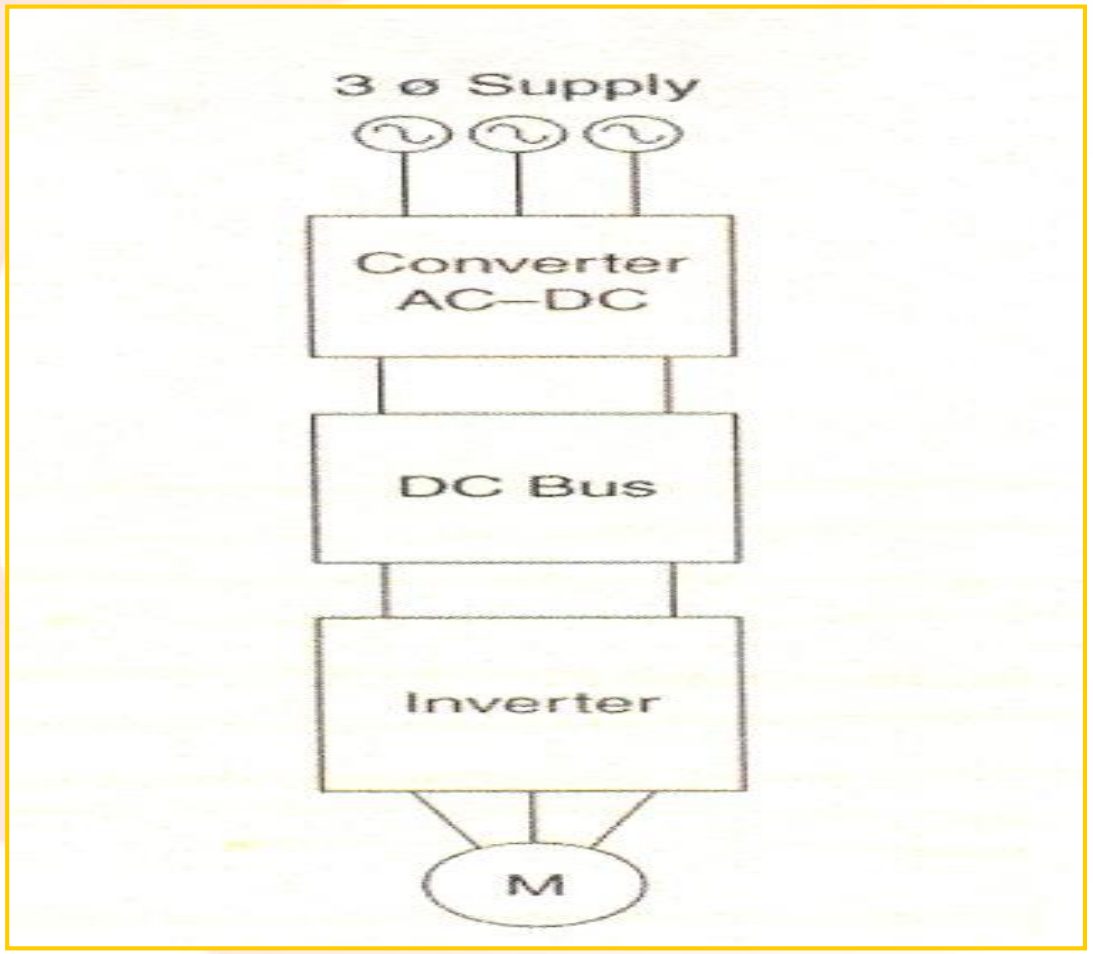
5) PLC Signals

Programmed Logic Controllers (PLC) coordinate the use of the AFD with numerous other schedules and commands. This can be used in very sophisticated control arrangements.



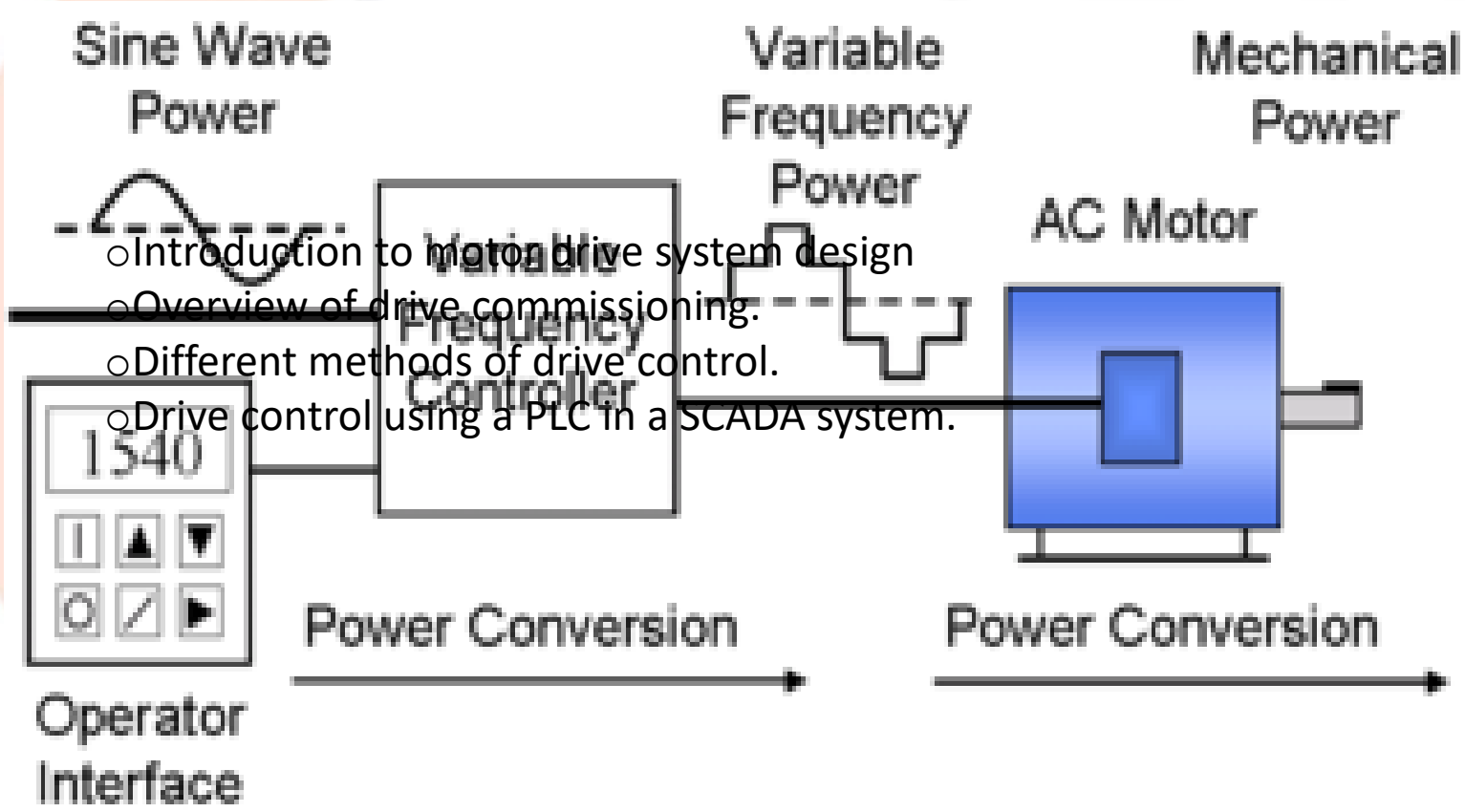


VFD POWER FLOW



VFD BLOCK DIAGRAM





- Introduction to motor drive system design
- Overview of drive commissioning.
- Different methods of drive control.
- Drive control using a PLC in a SCADA system.

What is inside a VFD ?

